

POLYP DETECTION MEDICAL REPORT

Analysis Date: June 04, 2026

VIDEO INFORMATION

Video ID:	808fdace-faf4-4d8e-9447-950e2fef30e8
Total Frames:	15016
Frame Rate:	25.0 fps (assumed)
Duration:	600.6 seconds

DETECTION SUMMARY

Detection Method	Count	Status
YOLO Detections	8997	✓
RT-DETR Detections	10557	✓
MedSAM2 Detections	10486	✓
Consensus (3-Model Agreement)	4	✓
Risk Classification	Count	Percentage
HIGH RISK	0	0.0%
MEDIUM RISK	3	75.0%
LOW RISK	1	25.0%
TOTAL ANALYZED	4	100.0%

DETAILED POLYP ANALYSIS

POLYP #1 — LIFTED_POLYP [LOW_RISK] (Tracked across 5522 frames)



Feature	Value	Interpretation
Frame Index	4710	Frame 4710 of 15016
Detection Method	Detector Consensus (MedSAM2 Unavailable)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 62.5%
Redness Score	0.075	Color-based vascularization
Relative Radius	29.9%	Polyp size as % of frame diagonal
Texture Score	0.060	Surface morphology
Vessel Visibility	0.005	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	NORMAL_MUCOSA	Normal mucosal pattern — no polyp morphology identified

Detection Confidence	89.4%	YOLO / RT-DETR / track confidence
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Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #2 — FLAT_POLYP [MEDIUM_RISK] (Tracked across 161 frames)

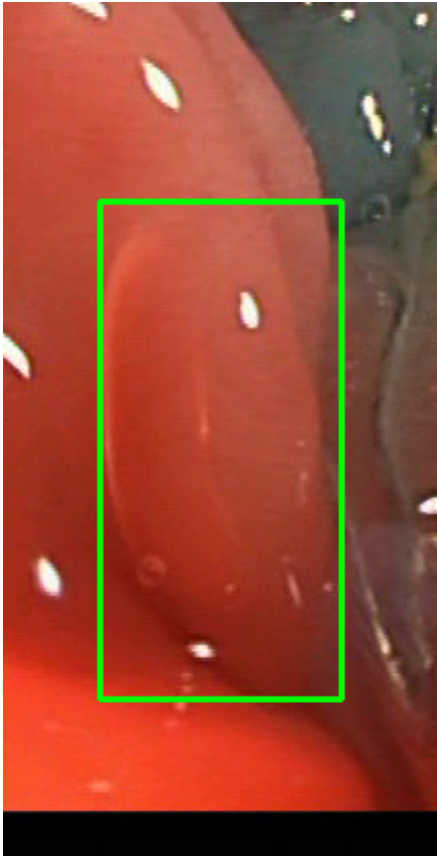


Feature	Value	Interpretation
Frame Index	12480	Frame 12480 of 15016
Detection Method	Detector Consensus (YOLO / RT-DETR / MedSAM2)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	FLAT_POLYP	Type confidence: 70.7%
Redness Score	0.091	Color-based vascularization
Relative Radius	20.2%	Polyp size as % of frame diagonal
Texture Score	0.191	Surface morphology
Vessel Visibility	0.038	Vascularization indicator
Risk Tier	MEDIUM RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	88.5%	YOLO / RT-DETR / track confidence

Type Assessment: Flat mucosal lesion — chromoendoscopy or NBI assessment recommended.

Low-confidence MODERATE RISK detection - Clinical review recommended

POLYP #3 — FLAT_POLYP [MEDIUM_RISK] (Tracked across 1114 frames)



Feature	Value	Interpretation
Frame Index	13469	Frame 13469 of 15016
Detection Method	Detector Consensus (MedSAM2, DETR)	Model Agreement (MedSAM2 unavailable for this video)
POLYP TYPE	FLAT_POLYP	Type confidence: 92.0%
Redness Score	0.156	Color-based vascularization
Relative Radius	22.1%	Polyp size as % of frame diagonal
Texture Score	0.164	Surface morphology
Vessel Visibility	0.432	Vascularization indicator
Risk Tier	MEDIUM RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output

Clinical Class	NORMAL_MUC OSA	Normal mucosal pattern — no polyp morphology identified
Detection Confidence	86.9%	YOLO / RT-DETR / track confidence

Type Assessment: Flat mucosal lesion — chromoendoscopy or NBI assessment recommended.

Low-confidence MODERATE RISK detection - Clinical review recommended

POLYP #4 — FLAT_POLYP [MEDIUM_RISK] (Tracked across 776 frames)



Feature	Value	Interpretation
Frame Index	7429	Frame 7429 of 15016
Detection Method	Detector Consensus (YOLO / RT-DETR)	Model Agreement (MedSAM2 unavailable for this video)
POLYP TYPE	FLAT_POLYP	Type confidence: 90.1%
Redness Score	0.115	Color-based vascularization
Relative Radius	32.4%	Polyp size as % of frame diagonal
Texture Score	0.036	Surface morphology
Vessel Visibility	0.646	Vascularization indicator
Risk Tier	MEDIUM RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	84.1%	YOLO / RT-DETR / track confidence

Type Assessment: Flat mucosal lesion — chromoendoscopy or NBI assessment recommended.

Low-confidence MODERATE RISK detection - Clinical review recommended

CLINICAL IMPRESSION

Summary:

A total of 4 polyps were detected and analyzed using multi-model consensus and AI-based risk stratification.

Risk Distribution:

- HIGH RISK polyps: 0 (0.0%)
- MEDIUM RISK polyps: 3 (75.0%)
- LOW RISK polyps: 1 (25.0%)

Recommendation:

The presence of MEDIUM RISK polyps warrants clinical review and correlation with endoscopic findings.

Technical Details:

- Detection: Multi-model consensus (YOLO, RT-DETR, MedSAM2)
- Features: 444-dimensional vectors (384 SSL + 60 biomarkers)
- Classification: Cluster-specific Decision Tree experts
- Confidence: Based on deep learning + medical heuristics

Disclaimer: This report is generated by an AI-assisted research system for academic and research purposes only. All clinical recommendations are derived from published ESGE 2017, NICE CG118/CG131, and BSG 2019 colonoscopy guidelines. This output does not constitute medical advice and must not be used as the sole basis for clinical decisions. All findings must be reviewed and confirmed by a qualified gastroenterologist or endoscopist.